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# BREATHING AND EXCHANGE OF GASES

## NCERT QUESTIONS WITH ANSWERS

**1. Define vital capacity. What is its significance?**

☞ Vital capacity is the volume of air that can breathe in after a forced expiration or Volume of air that can breathe out after a forced inspiration.
**Significance:** Greater the vital capacity more is the energy available to the body.

**2. State the volume of air remaining in the lungs after a normal breathing.**

☞ Functional residual capacity (FRC). FRC= RV + ERV = 2200 – 2300 ml.

**3. Diffusion of gases occurs in the alveolar region only and not in the other parts of respiratory system. Why?**

☞ For effective gas exchange, respiratory surface must be thin, highly vascular and with large surface area. Only alveolar region has these features.

**4. What are the major transport mechanisms for $CO_2$? Explain.**

☞ **$CO_2$ transport** occurs in 3 ways:
- **As carbonic acid:** In tissues, about **7%** of $CO_2$ is dissolved in **plasma water** to form **carbonic acid** and carried to lungs.
- **As carbamino-haemoglobin:** In tissues, **20-25%** of $CO_2$ binds to Hb to form **carbamino-haemoglobin**. In alveoli, $CO_2$ dissociates from carbamino-haemoglobin.
- **As bicarbonates:** About **70%** of $CO_2$ is transported by this method. RBCs and plasma contain an enzyme, *carbonic anhydrase*. This enzyme facilitates the following reactions.

$$CO_2 + H_2O \xrightleftharpoons{\text{Carbonic anhydrase}} H_2CO_3$$
$$H_2CO_3 \xrightleftharpoons{\text{Carbonic anhydrase}} HCO_3^- + H^+$$

**5. What will be the $pO_2$ and $pCO_2$ in the atmospheric air compared to those in the alveolar air?**

**(i) $pO_2$ lesser, $pCO_2$ higher**

**(ii) $pO_2$ higher, $pCO_2$ lesser**

**(iii) $pO_2$ higher, $pCO_2$ higher**

**(iv) $pO_2$ lesser, $pCO_2$ lesser.**

☞ $pO_2$ higher, $pCO_2$ lesser.

**6. Explain the process of inspiration under normal conditions.**

☞ During inspiration, the **diaphragm** contracts (flattens) causing an increase in vertical volume ***(antero-posterior axis).***
Contraction of **external intercostal muscles** (muscles found between ribs) lifts up the ribs and sternum causing an increase in thoracic volume in the ***dorso-ventral axis***.
These changes reduce pressure inside the thorax causing the expansion of lungs. Thus pulmonary volume increases resulting in decrease of ***intra-pulmonary pressure*** to less than the atmospheric pressure. So air moves into lungs.

**7. How is respiration regulated?**

☞ Respiratory centres present in the brain include
- **Respiratory rhythm centre (Inspiratory & Expiratory centres):** In **medulla oblongata**.
- **Pneumotaxic centre:** In **Pons**. It moderates the functions of the respiratory rhythm centre. Neural signal from this centre reduces the duration of inspiration and thereby alter the respiratory rate.
- **Chemosensitive area:** Seen adjacent to the rhythm centre. Increase in the concentration of $CO_2$ and $H^+$ activates this centre, which in turn signals rhythm centre. **Receptors** in **aortic arch** and **carotid artery** also recognize changes in $CO_2$ and $H^+$ concentration and send signals to the rhythm centre.

**8. What is the effect of $pCO_2$ on oxygen transport?**

☞ Increase in $pCO_2$ in blood causes rightward shift of the oxygen-haemoglobin dissociation curve. It decreases the affinity of Hb for oxygen (Bohr's effect). It plays a crucial role in enhancing oxygenation of blood in the lungs and release of oxygen in the tissues.

**9. What happens to the respiratory process in a man going up a hill?**

☞ His breathing rate will increase.

**10. What is the site of gaseous exchange in an insect?**

☞ Tracheae.

**11. Define oxygen dissociation curve. Can you suggest any reason for its sigmoidal pattern?**

☞ Oxygen dissociation curve is a sigmoid curve obtained when percentage saturation of Hb with $O_2$ is plotted against the $pO_2$.

The sigmoidal pattern is due to following properties:

a) Minimal loss of $O_2$ from haemoglobin occurs above $pO_2$ of 70-80 mm Hg despite significant changes in tension of oxygen beyond this. This is depicted by flat portion of the curve.
b) Significant change of the dissociation curve below $pO_2$ of 40 mm Hg ensures that with any further decline in $pO_2$ there will be a greater release of oxygen from haemoglobin.

**12. Have you heard about hypoxia? Try to gather information about it, and discuss with your friends.**

☞ Hypoxia is a condition of oxygen shortage in the tissues. It is 2 types:
**(i) Artificial hypoxia:** It is due to shortage of oxygen in the air as at high altitude. It causes mountain sickness. Its symptoms are breathlessness, headache, dizziness and bluish tinge on skin.
**(ii) Anaemic hypoxia:** It is due to the reduced oxygen carrying capacity of the blood due to anaemia or carbon monoxide poisoning. Here, less haemoglobin is available for carrying $O_2$.

**13. Distinguish between**

**(a) IRV and ERV**

**(b) Inspiratory capacity and Expiratory capacity.**

**(c) Vital capacity and Total lung capacity.**

☞ **(a) Inspiratory reserve volume (IRV)** is an additional volume of air that can inspire by forceful inspiration.
**Expiratory reserve volume (ERV)** is an additional volume of air that can expire by a forceful expiration.
**(b) Inspiratory capacity (IC)** is the total volume of air inspired after a normal expiration (TV + IRV).
**Expiratory capacity (EC)** is the total volume of air expired after a normal inspiration (TV + ERV).
**(c) Vital capacity (VC)** is the volume of air that can breathe in after a forced expiration or Volume of air that can breathe out after a forced inspiration (ERV + TV + IRV).
**Total lung capacity (TLC)** is the total volume of air in the lungs after a maximum inspiration. (RV + ERV + TV + IRV or VC + RV).

**14. What is Tidal volume? Find out the Tidal volume (approximate value) for a healthy human in an hour.**

☞ **Tidal volume (TV)** is the volume of air inspired or expired during a normal respiration. It is about **500 ml.**
TV for a healthy human per minute = 500 x breathing rate (12-16) = 6000 to 8000 ml.
Hence TV in an hour = 6000 to 8000 x 60 = 360000 – 480000 ml.